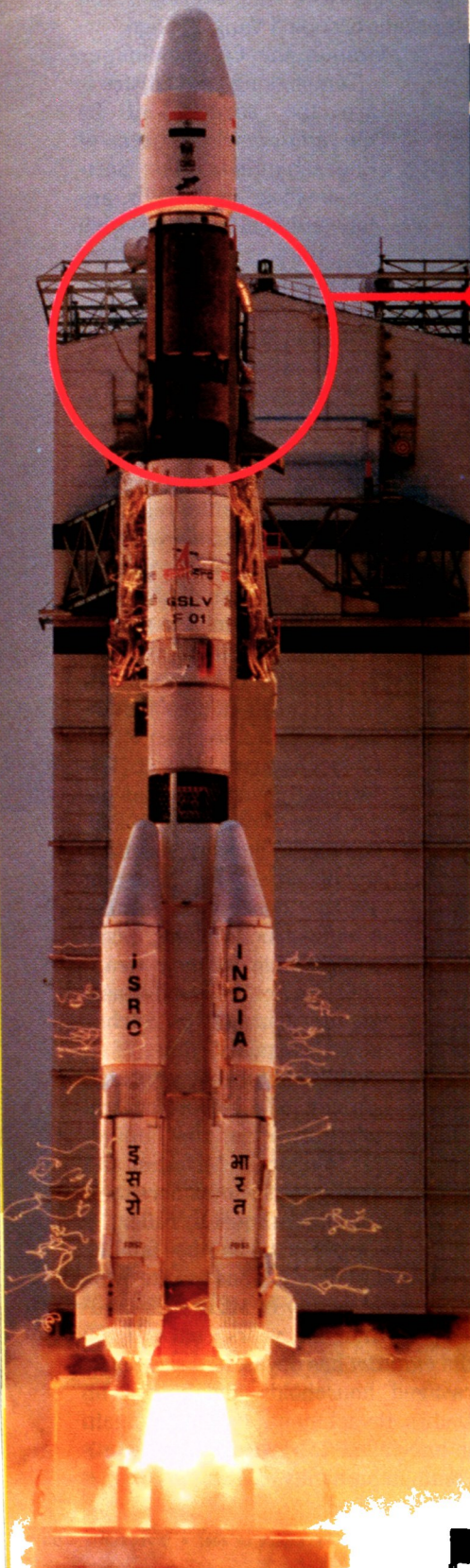


JUST SUPERCOOL

ISRO overcomes major hurdles to test an indigenous cryogenic engine giving it mastery over the entire range of space rocketry



BIG BOOST: The cryogenic engine under test; (left) the GSLV that it would be used for

■ By Raj Chengappa

Armies tend to herald their victories with the blowing of bugles. The Indian Space Research Organisation (ISRO) chose to announce its entry into the exclusive club of nations capable of building cryogenic engines with a thunderous sound of a rocket test that lasted 12 long minutes. The sound, equivalent to that of 10 Jumbo jets taking off simultaneously, reverberated across the Mahendragiri hills in Tamil Nadu where India's ultra-modern liquid engine test facility is located.

It signalled the culmination of ISRO's 13-year quest to build engines using cryogenic fuels that are super-cooled to around minus 250 degrees. (Cryo originates from the Greek word Kryos which literally means ice-cold.) The quest began in 1993 when Russia, under pressure from the US, reneged from an agreement to supply India with the technology to build such engines. The Americans successfully argued that it would give India the capability to launch Intercontinental Ballistic Missiles (ICBMs).

ISRO needed cryogenic engines to power the upper stages of its Geo-Synchronous Launch Vehicle (GSLV)

which is capable of putting two-tonne INSAT-class communications satellite into a geo-synchronous orbit at 36,000 km above the earth. Cryogenic engines that use a mixture of super-cooled liquid hydrogen and oxygen as propellants are highly efficient providing more 'bounce per ounce' of fuel. The thrust these engines generate is usually 50 per cent more than that provided either by liquid or solid rockets for the same weight of fuel (*see graphic*). In doing so, it gives a space-faring nation the flexibility to put heavier satellites or other payloads into space.

So as not to delay its GSLV programme, ISRO renegotiated the agreement with Russia for outright purchase of two cryogenic engines. It simultaneously began designing and building an indigenous cryogenic engine similar to the Russian one for future flights. ISRO had hoped to complete the indigenous project within five years but the sheer complexity of building such an engine frustrated scientists for many years. Barring the performance designs given by the Russians, the fully assembled engines purchased by them were, as one scientist put it, "like a blackbox". That meant that ISRO had to do everything from scratch. Given that the rocket has